COHOMOLOGY ON A COMPONENT AGREES WITH COHOMOLOGY ON THE TOTAL SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: in-source stubs lemma-simplicial-site-site, lemma-simplicial-cocontinuous-site, lemma-restriction-injective-to-component-site, Situation situation-simplicial-site.

Editorial note. The label lemma-sanity-check names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Total site construction, cases (A) and (B). Same unit, Lemma lemma-simplicial-site-site and Lemma lemma-simplicial-cocontinuous-site. the two constructions of C_total that give 'U viewed as an object of C_total' its precise meaning.
- Localization invariance of cohomology. Same unit, Lemma lemma-restriction-injective-to-component-site. the equality of cohomology under the localization functor C_n -> C used for the final equality.

Source reference. Original reference 09WH: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1086–1086. Prerequisite pointer invoked at proof line(s) 1897; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WD: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 844–844. Prerequisite pointer invoked at proof line(s) 1895; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WC: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 797–797. Prerequisite pointer invoked at proof line(s) 1894; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 09WE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 888–888. Prerequisite pointer invoked at proof line(s) 1884; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\mathcal{C}$ be as in Situation 09WE. Let $U \in \mathrm{Ob}(\mathcal{C}_n)$. Let $\mathcal{F} \in Ab(\mathcal{C}_{total})$. Then $H^p(U, \mathcal{F}) = H^p(U, g_n^{-1}\mathcal{F})$ where on the left hand side U is viewed as an object of $\mathcal{C}_{total}$.*

Proof. Observe that "U viewed as object of $\mathcal{C}_{total}$" is explained by the construction of $\mathcal{C}_{total}$ in Lemma 09WC in case (A) and Lemma 09WD in case (B). The equality then follows from Lemma 09WH and the definition of cohomology. □